

Metadata Record: iNaturalist Research-Grade Observations Retrieved via GBIF

Dataset Title

iNaturalist Research-Grade Observations (GBIF-mediated download)

Dataset Description

This dataset consists of research-grade iNaturalist observations distributed through the Global Biodiversity Information Facility (GBIF). The download was requested in Darwin Core Archive (DwC-A) format using the GBIF Occurrence Download API and constrained by geographic, taxonomic, and administrative filters.

The source dataset is the GBIF dataset **iNaturalist Research-grade Observations** (`datasetKey = 50c9509d-22c7-4a22-a47d-8c48425ef4a7`), published by iNaturalist.org and contributed by iNaturalist users.

Data Procurement

Source

- Provider: Global Biodiversity Information Facility (GBIF)
- API Endpoint:
 - <https://api.gbif.org/v1/occurrence/download/request>
- Download Format:
 - Darwin Core Archive (DWCA)

Download Query Payload

```
{
  "format": "DWCA",
  "predicate": {
    "type": "and",
    "predicates": [
      {
        "type": "equals",
        "key": "COUNTRY",
        "value": "US"
      },
      {
        "type": "equals",
        "key": "STATE_PROVINCE",
        "value": "North Carolina"
      },
      {
        "type": "equals",
        "key": "DATASET_KEY",
        "value": "50c9509d-22c7-4a22-a47d-8c48425ef4a7"
      },
      {
        "type": "equals",
        "key": "TAXON_KEY",
        "value": "212"
      },
      {
        "type": "within",
        "geometry": "POLYGON((-79.312 35.866,-78.458 35.866,-78.458 36.267,-79.312 36.267,-79.312 35.866))"
      }
    ]
  }
}
```

Query Interpretation

The download includes records meeting all of the following criteria:

Filter	Value
Country	United States
State/Province	North Carolina
Dataset	iNaturalist Research-Grade Observations
GBIF Taxon Key	212
Spatial Extent	Polygon bounding a portion of central North Carolina (Durham–Orange–Person County region)
Record Type	Research-grade iNaturalist observations distributed through GBIF

Spatial Extent

The download is restricted to observations falling within the polygon:

```
POLYGON(  
  (-79.312 35.866,  
   -78.458 35.866,  
   -78.458 36.267,  
   -79.312 36.267,  
   -79.312 35.866)  
)
```

Coordinate Reference System: WGS84 (EPSG:4326)

Taxonomic Scope

The query restricts records to:

- GBIF Taxon Key: **212**

Users should consult the GBIF Backbone Taxonomy corresponding to the download date to determine the exact taxonomic concept represented by this taxon key.

Data Format

The dataset was downloaded as a **Darwin Core Archive (DwC-A)**.

A DwC-A download typically contains:

File	Description
occurrence.txt	Core occurrence records
multimedia.txt	Associated media (images, audio)
meta.xml	Archive schema definition
eml.xml	Dataset metadata
rights.txt	Licensing information (when present)

Expected Content

The occurrence table generally contains biodiversity occurrence records described using Darwin Core standards.

Common fields include:

Record Identifiers

- gbifID
- occurrenceID
- datasetKey
- institutionCode
- collectionCode

Taxonomy

- kingdom
- phylum
- class
- order
- family
- genus
- species
- scientificName
- acceptedScientificName
- taxonRank
- taxonKey

Observation Information

- basisOfRecord
- occurrenceStatus
- eventDate
- year
- month
- day
- recordedBy
- identifiedBy

Location Information

- decimalLatitude
- decimalLongitude
- coordinateUncertaintyInMeters
- countryCode
- country
- stateProvince
- county
- locality

Data Quality and Provenance

- issue
- publishingOrgKey
- datasetName
- license
- rightsHolder
- informationWithheld
- dataGeneralizations

Media-Related Fields

(typically in multimedia.txt)

- identifier
- type
- format
- references
- license
- creator

Data Quality Notes

The source dataset consists of **research-grade iNaturalist observations**, meaning observations have typically:

1. Media evidence (photo or sound).
2. A community-supported taxonomic identification.
3. Date information.
4. Geographic coordinates.
5. No unresolved quality issues preventing research-grade status.

Sensitive taxa may have obscured or generalized coordinates.

Recommended Citation

GBIF-mediated datasets should be cited using the DOI generated for the specific download when available. In addition, the source dataset should be acknowledged:

iNaturalist contributors. iNaturalist Research-grade Observations. Published by iNaturalist.org and accessed through the Global Biodiversity Information Facility (GBIF). Dataset key: 50c9509d-22c7-4a22-a47d-8c48425ef4a7.

Source Dataset

- Dataset Name: iNaturalist Research-grade Observations

- Dataset Key: 50c9509d-22c7-4a22-a47d-8c48425ef4a7
- Publisher: iNaturalist.org
- Access Platform: GBIF
- Format: Darwin Core Archive (DwC-A)

Metadata Creation

This metadata document was generated to accompany a GBIF occurrence download created from a JSON predicate query submitted to the GBIF Occurrence Download API.